Lecture 4

ChE 467/596-012
Polymer Rheology

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Last Class-Material Functions

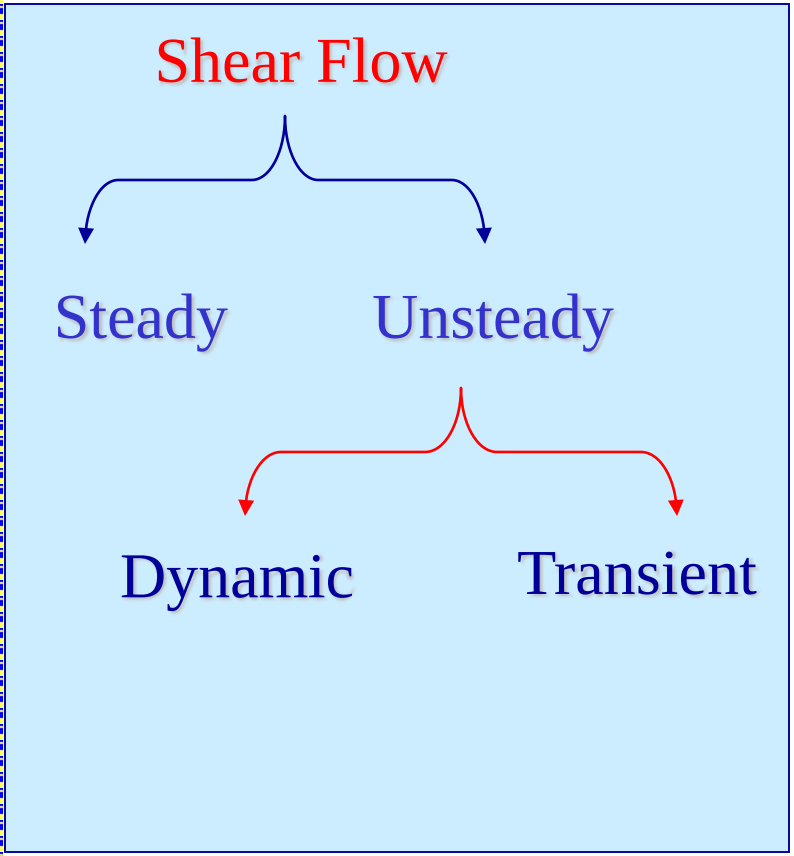
Inherent properties of materials

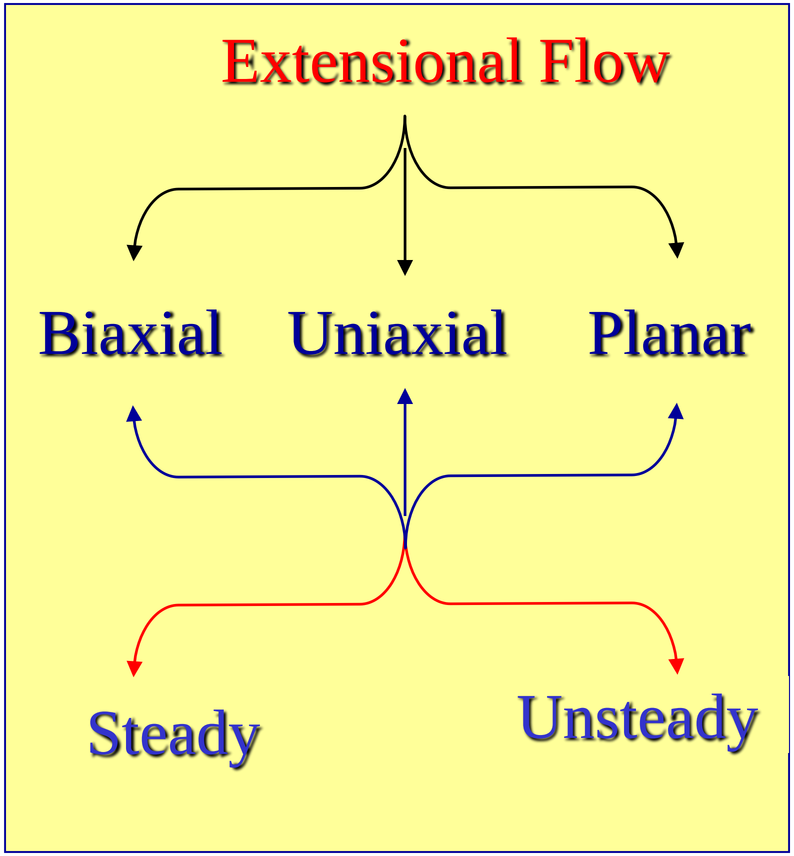
Serves to classify material

Used in developing constitution equations

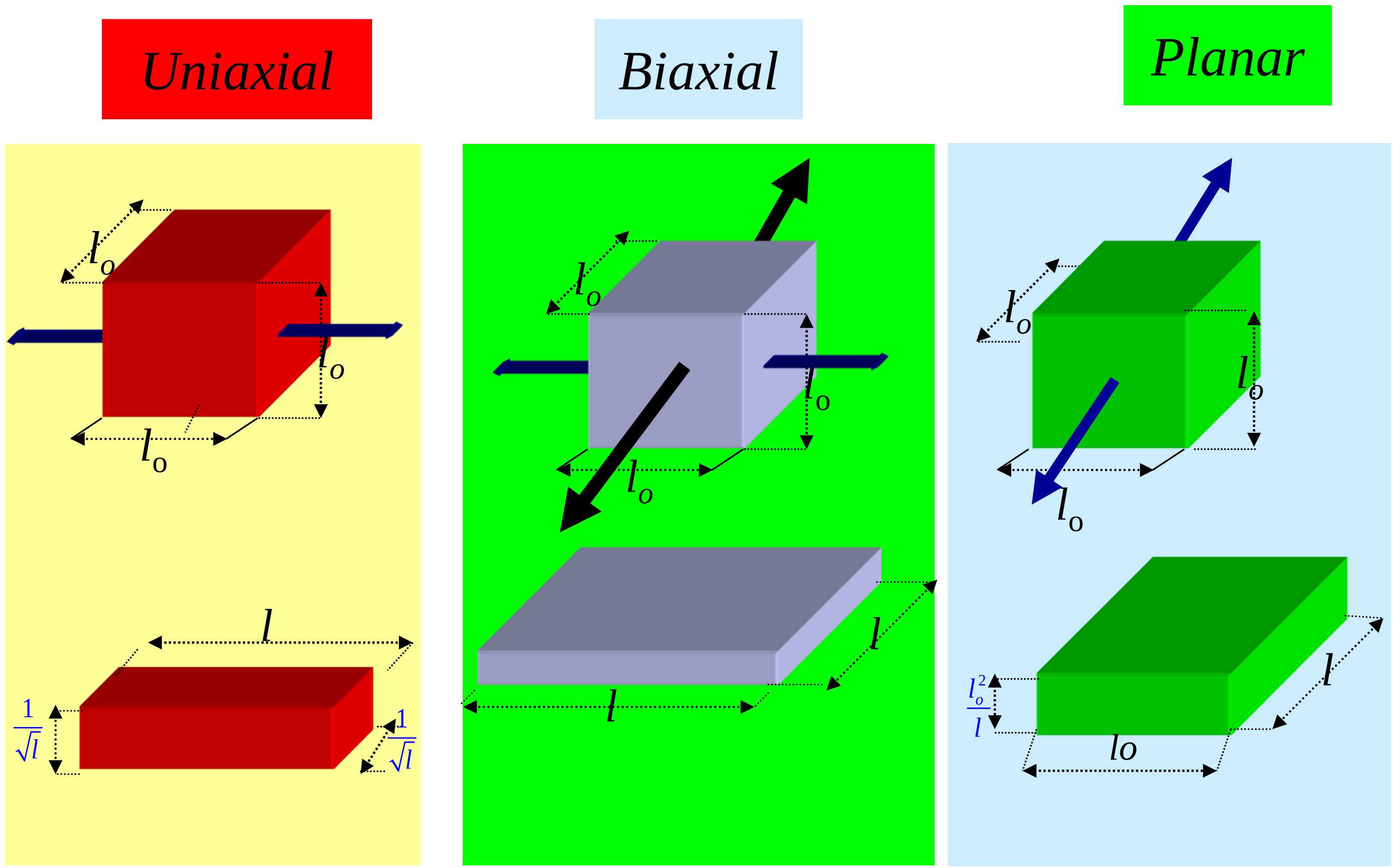
Provides information – microstructural or molecular level

Flow Type

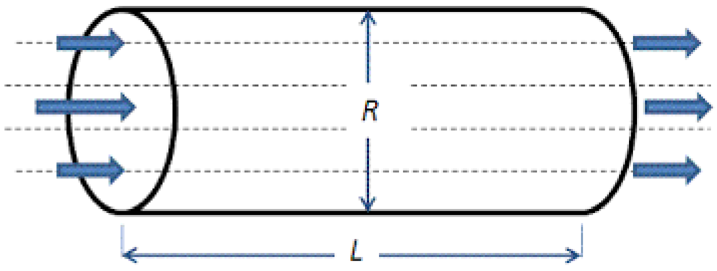

Shear Flow
Steady
Unsteady
Dynamic
Transient


Extensional Flow
Biaxial
Uniaxial
Planar
Steady
Unsteady

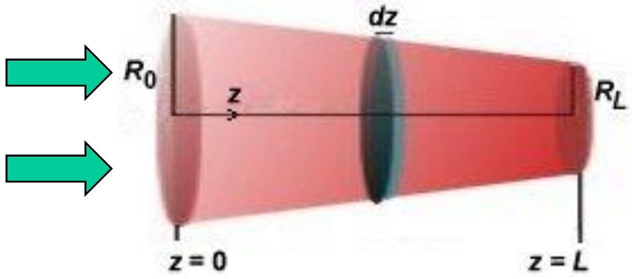
Extensional Flow



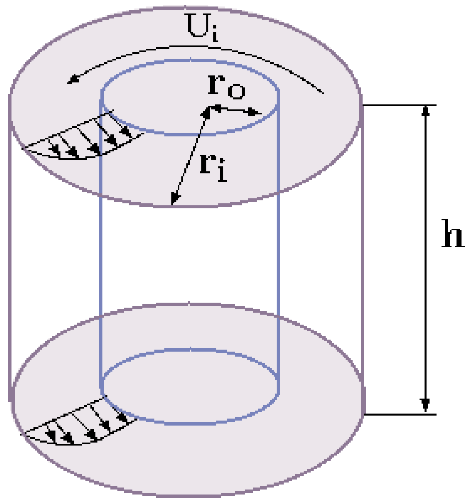
What kind of flow are these?



Flow through pipe



Flow through tapered pipe



Flow between concentric cylinders

Thus Far…..Moving Forward

- Flow Types: Shear and Extension
- Concept of Tensors: Stress and Deformation

Today

- A bit more on $\underline{\underline{T}} = -p\underline{\underline{I}} + \underline{\underline{\tau}}$
- Continue on Chap 3 DPL

Stress tensors in different flows

Extension :

$$\underline{\underline{T}} = \begin{bmatrix} -p+\tau_{xx} & 0 & 0 \\ 0 & -p+\tau_{yy} & 0 \\ 0 & 0 & -p+\tau_{zz} \end{bmatrix}$$

Shear:

$$\underline{\underline{T}} = \begin{bmatrix} -p+\tau_{xx} & \tau_{yx} & 0 \\ \tau_{xy} & -p+\tau_{yy} & 0 \\ 0 & 0 & -p+\tau_{zz} \end{bmatrix}$$

$$\tau_{xz} = \tau_{yz} = 0$$

i.e., only one shear component non-zero

For either flow,

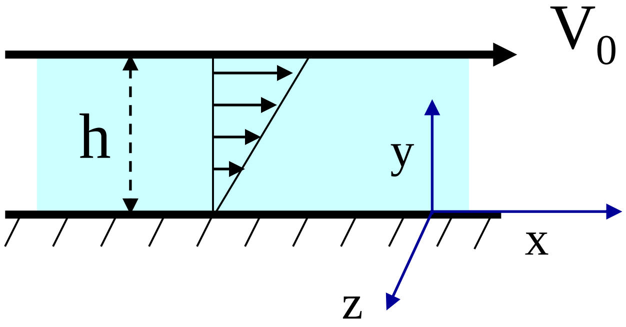
normal stresses of interest

$$T_{xx} - T_{yy} = \tau_{xx} - \tau_{yy}$$

$$T_{yy} - T_{zz} = \tau_{yy} - \tau_{zz}$$

Material function- steady shear flow

3 material functions characterize a material in a shear flow



$$v_x = \frac{V_o}{h} y = \dot{\gamma}\, y$$

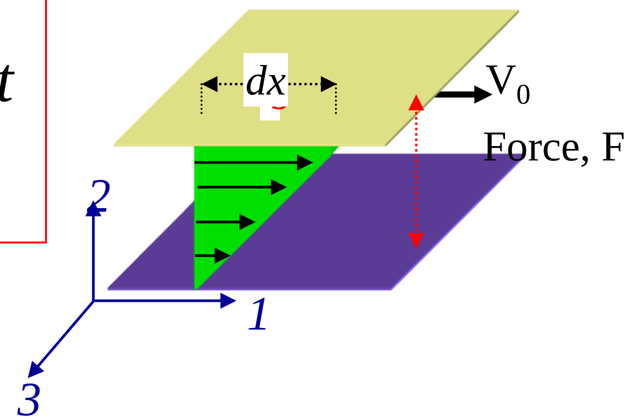
$$\eta(\dot{\gamma}) = \frac{\tau_{yx}}{\dot{\gamma}}$$ Viscosity

$$\psi_1(\dot{\gamma}) = \frac{\tau_{xx} - \tau_{yy}}{\dot{\gamma}^2} = \frac{N_1}{\dot{\gamma}^2}$$ First Normal Stress Coefficent

$$\psi_2(\dot{\gamma}) = \frac{\tau_{yy} - \tau_{zz}}{\dot{\gamma}^2} = \frac{N_2}{\dot{\gamma}^2}$$ Second Normal Stress Coefficent

Steady Shear Flow

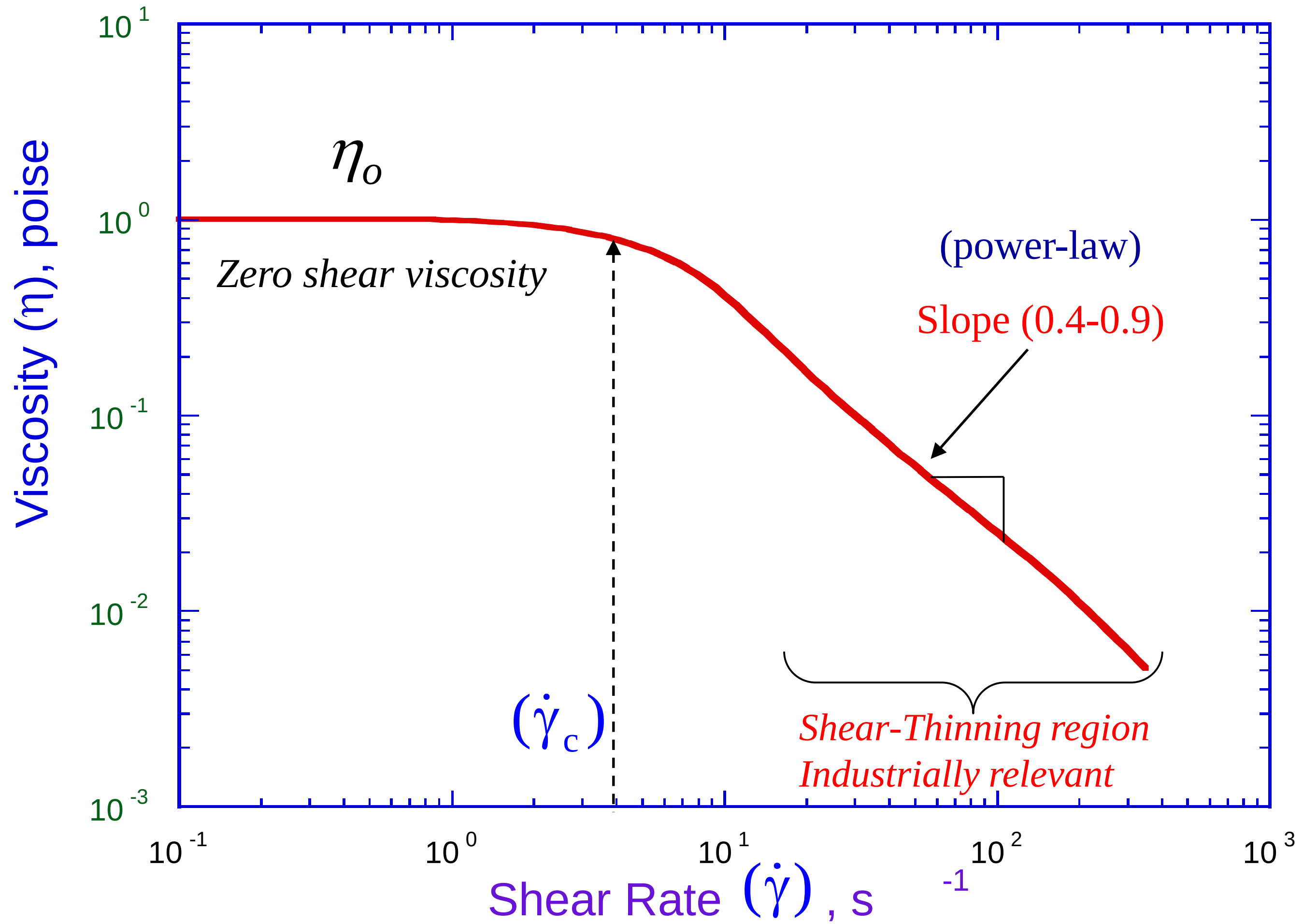
- *x(1) Flow Direction*
- *y(2) Direction of Velocity Gradient*
- *Z(3) Neutral Direction*



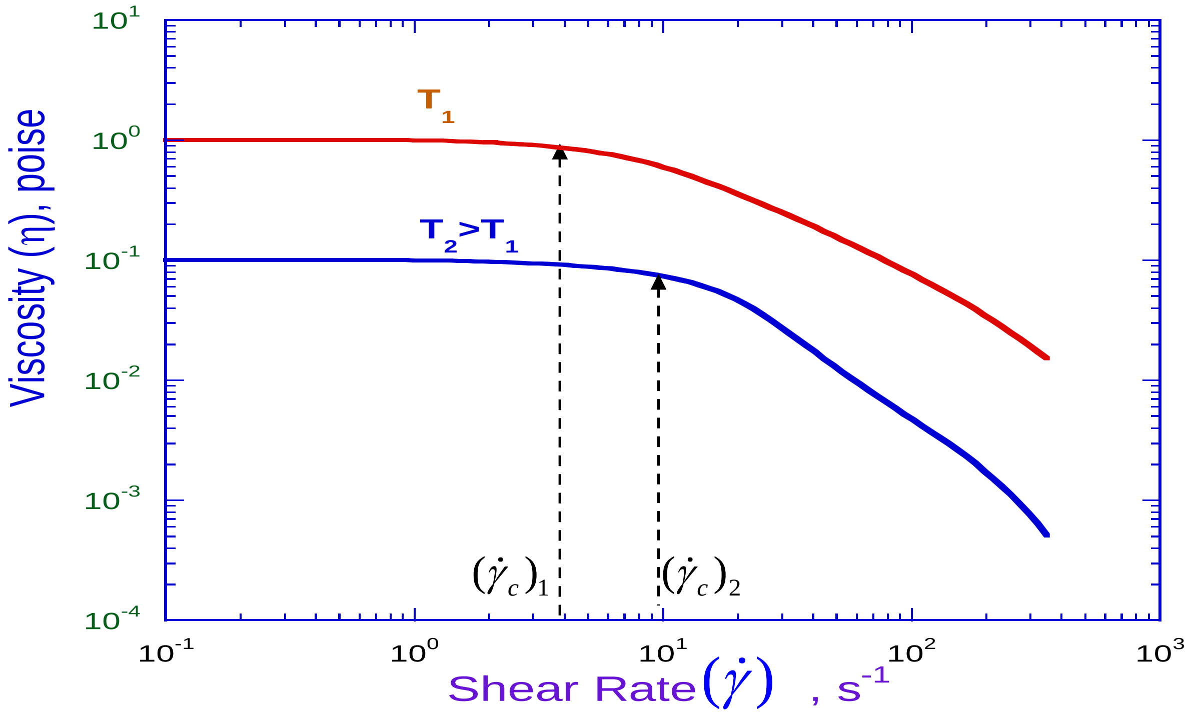
3 Stresses of Interest in Steady Shear Flow

- τ_{yx} Shear Stress
- $N_1 = \tau_{xx} - \tau_{yy}$ First Normal Stress Difference
- $N_2 = \tau_{yy} - \tau_{zz}$ Second Normal Stress Difference

Shear Viscosity



Temperature effect

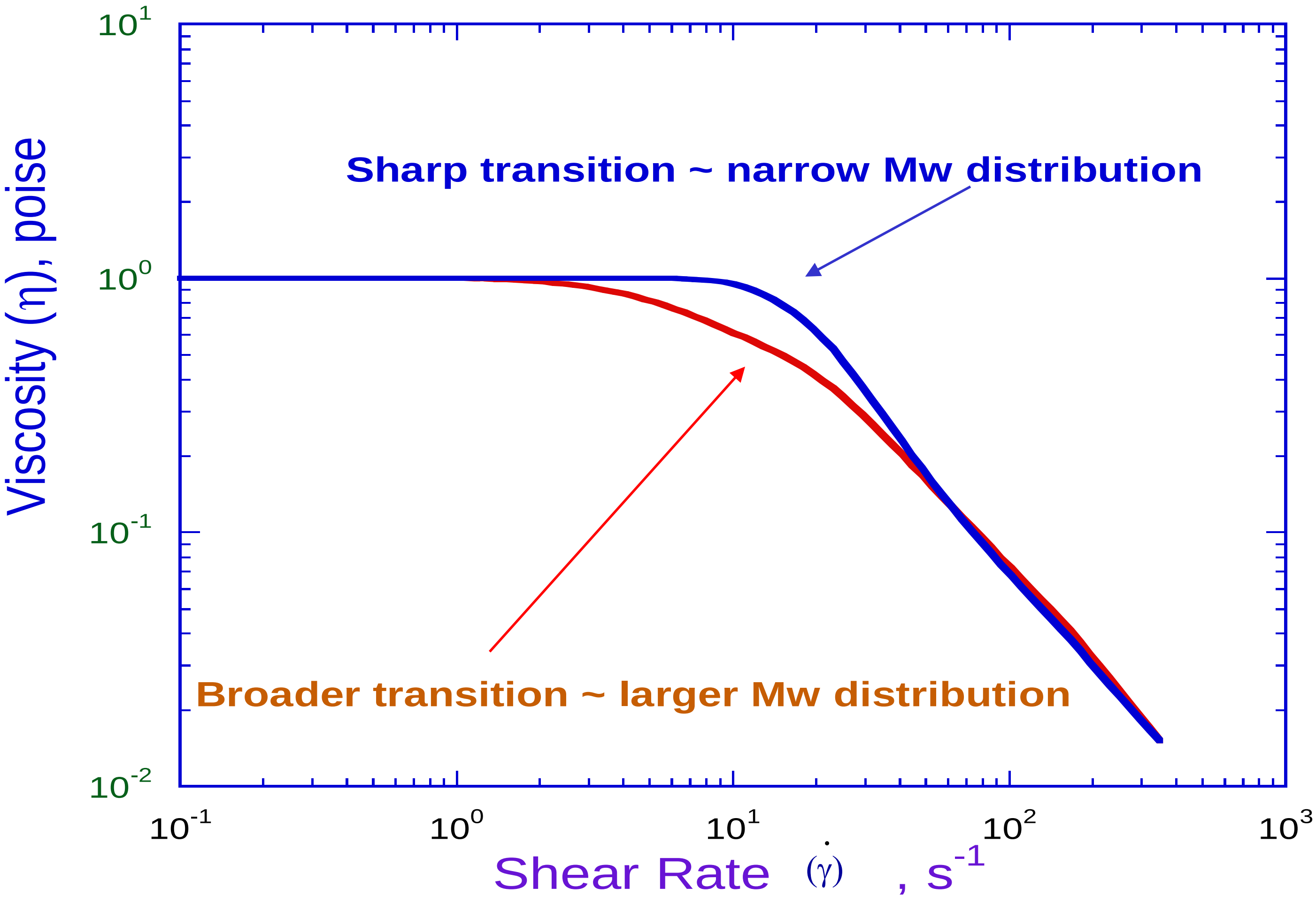


As Temperature ↑

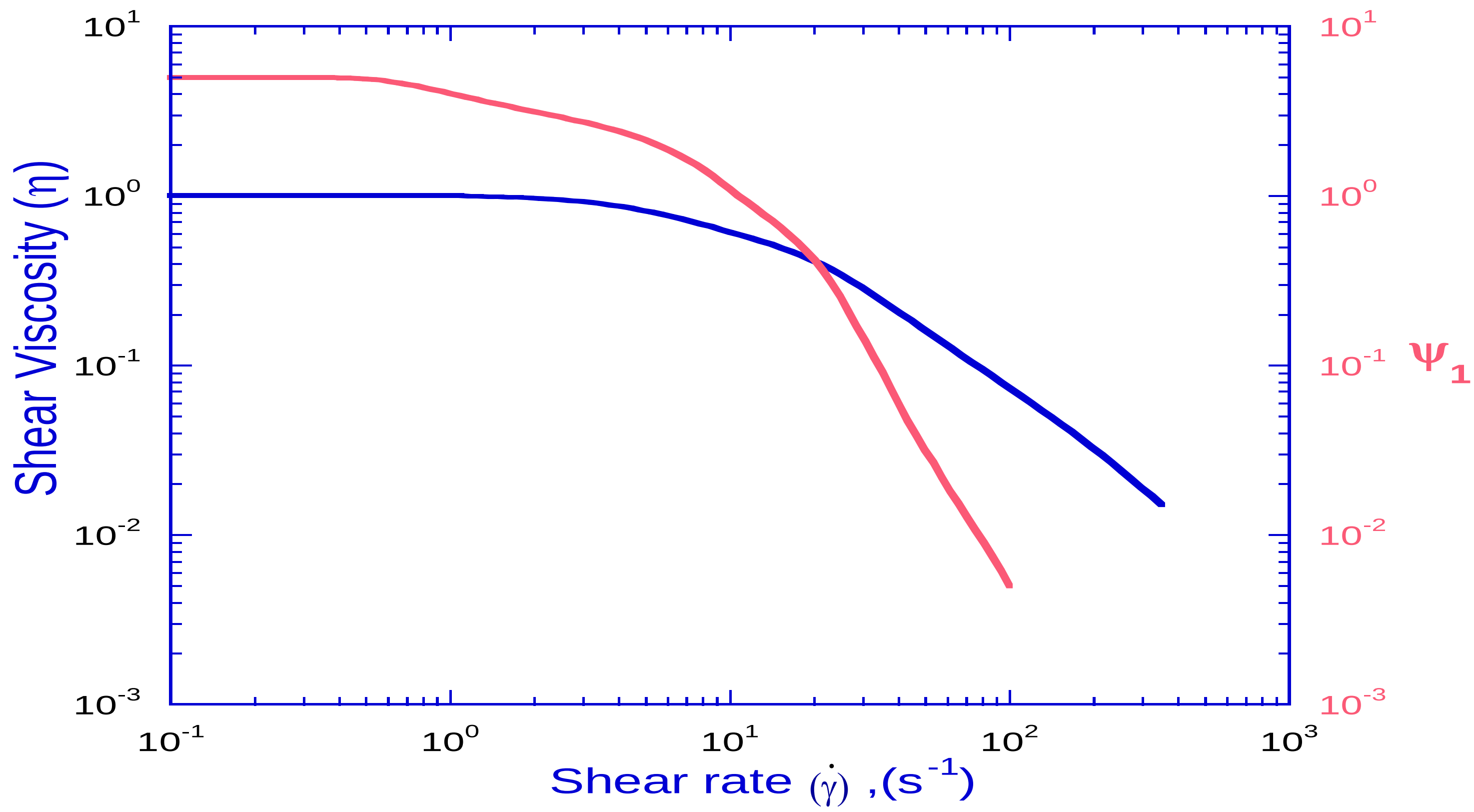
η_o ↓ and $(\dot{\gamma}_c)$↑

λ ↓ (Longest relaxation time of material $\cong 1/(\dot{\gamma}_c)$)

Molecular Weight Distribution



η versus Normal Stress Coefficient



Slope of normal Stress Coefficient is steeper than that of viscosity

Class Exercise: Identifying Flows/Fluids

Given: Newtonian fluid

$$\underline{\underline{T}} = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

What flow type?

$$\underline{\underline{T}} = \begin{bmatrix} 9 & 8 & 0 \\ 8 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

What flow type?

Given: Non-Newtonian fluid

$$\underline{\underline{T}} = \begin{bmatrix} 9 & 8 & 0 \\ 8 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

What flow type?

No information available

$$\underline{\underline{T}} = \begin{bmatrix} 18 & 7 & 0 \\ 7 & 18 & 0 \\ 0 & 0 & 18 \end{bmatrix}$$

Flow type?
Fluid type?

Newtonian

Class Exercise: Identifying Flows/Fluids

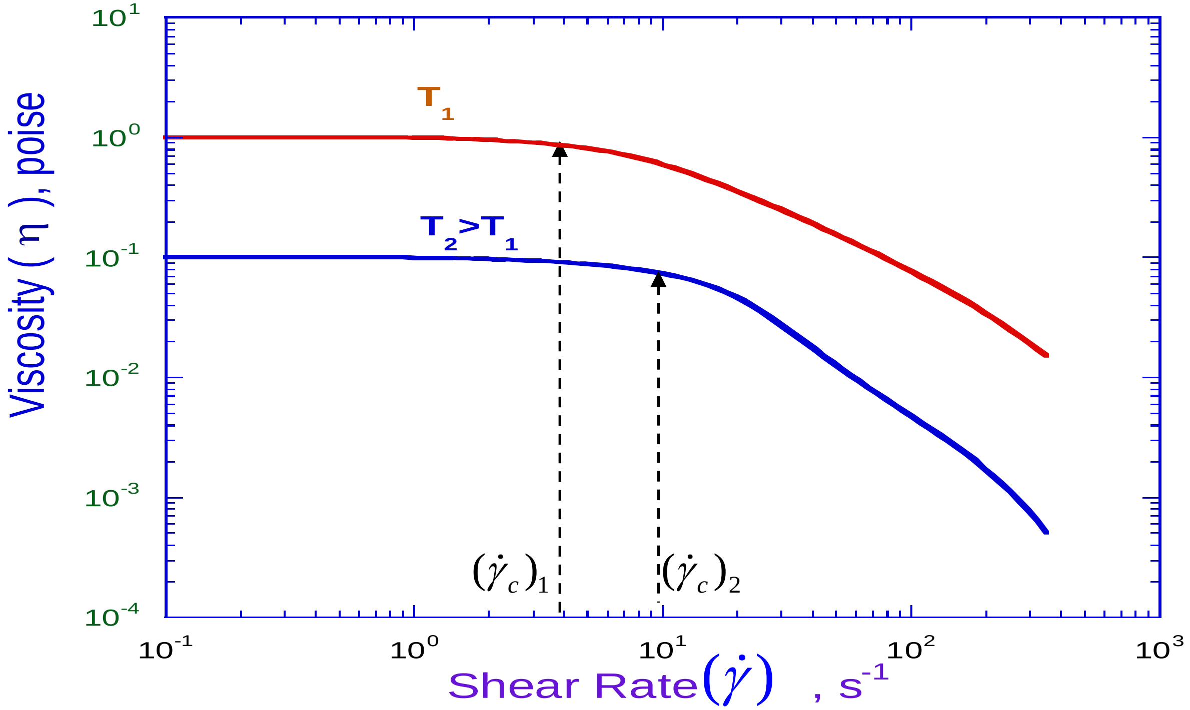
No information available

$$\underline{\underline{T}} = \begin{bmatrix} 18 & 7 & 0 \\ 7 & 18 & 0 \\ 0 & 0 & 18 \end{bmatrix}$$

Flow type?
Fluid type?

Newtonian

Temperature effect

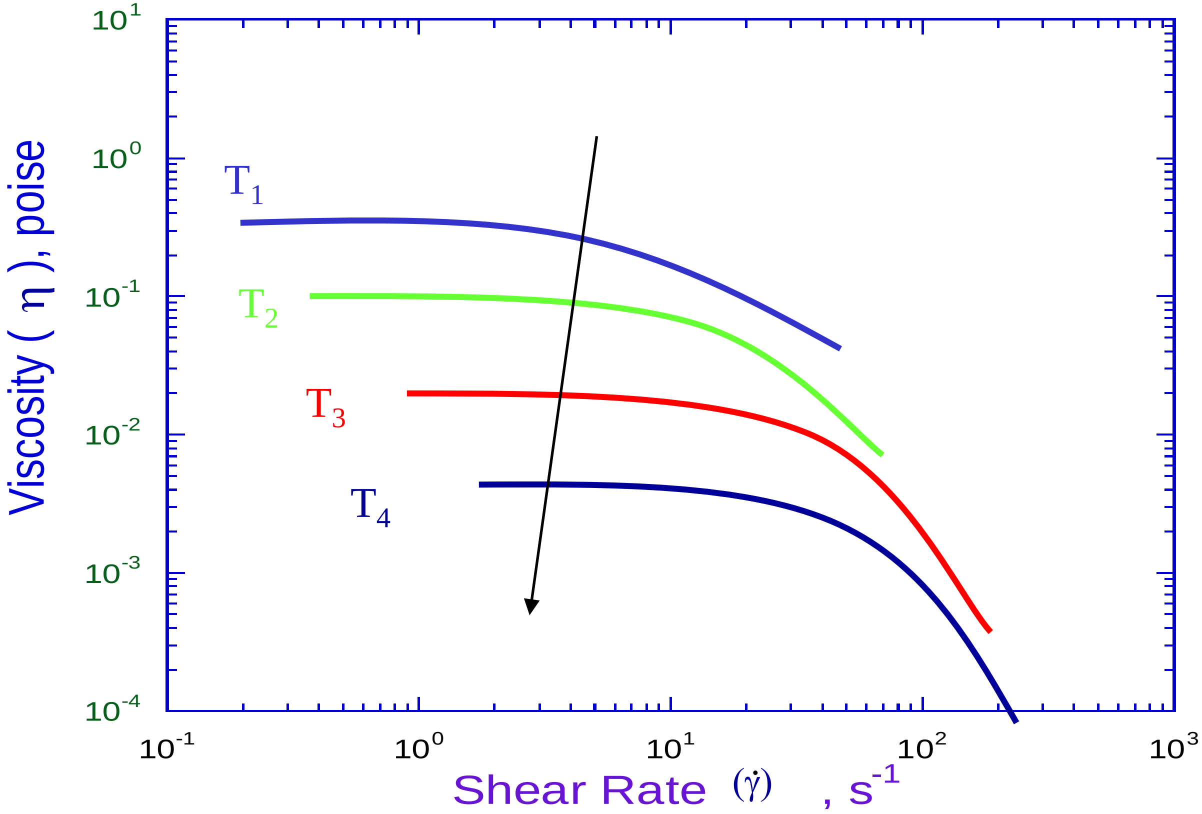


As Temperature ↑

η_o ↓ and $(\dot{\gamma}_c)$ ↑

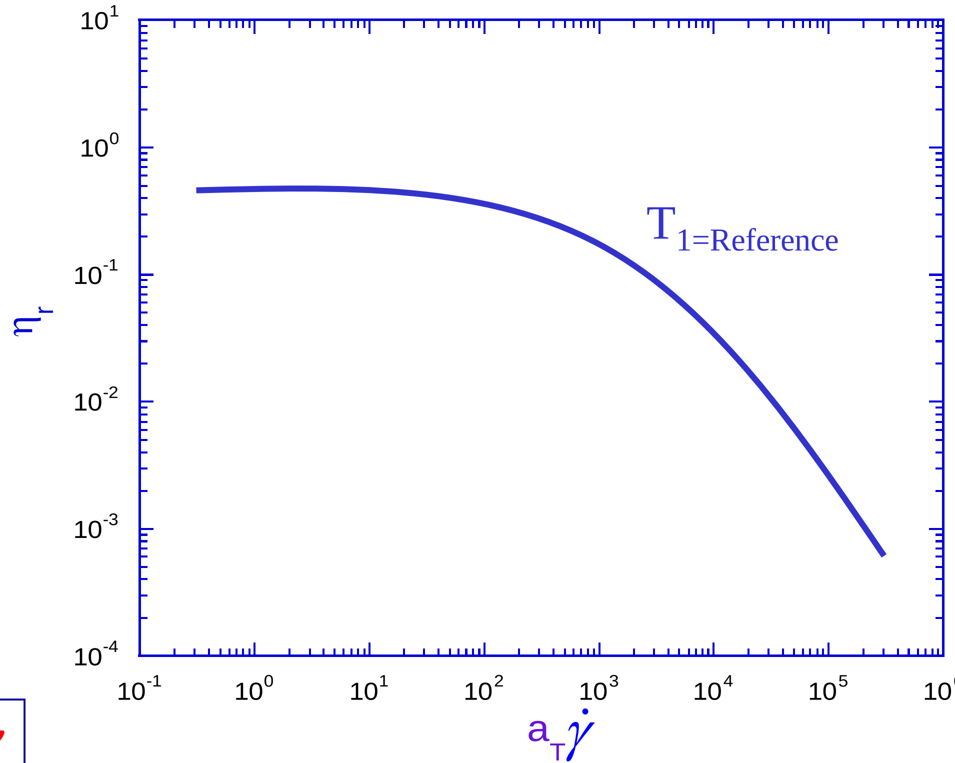
λ ↓ (Longest relaxation time of material ≅ $1/(\dot{\gamma}_c)$)

Time-Temperature Superposition (TTS)



η at temp T at shear rate $\dot{\gamma}$ is equivalent to viscosity at some reference temperature measured at shear rate $a_T \dot{\gamma}$

TTS –also method of reduced variables



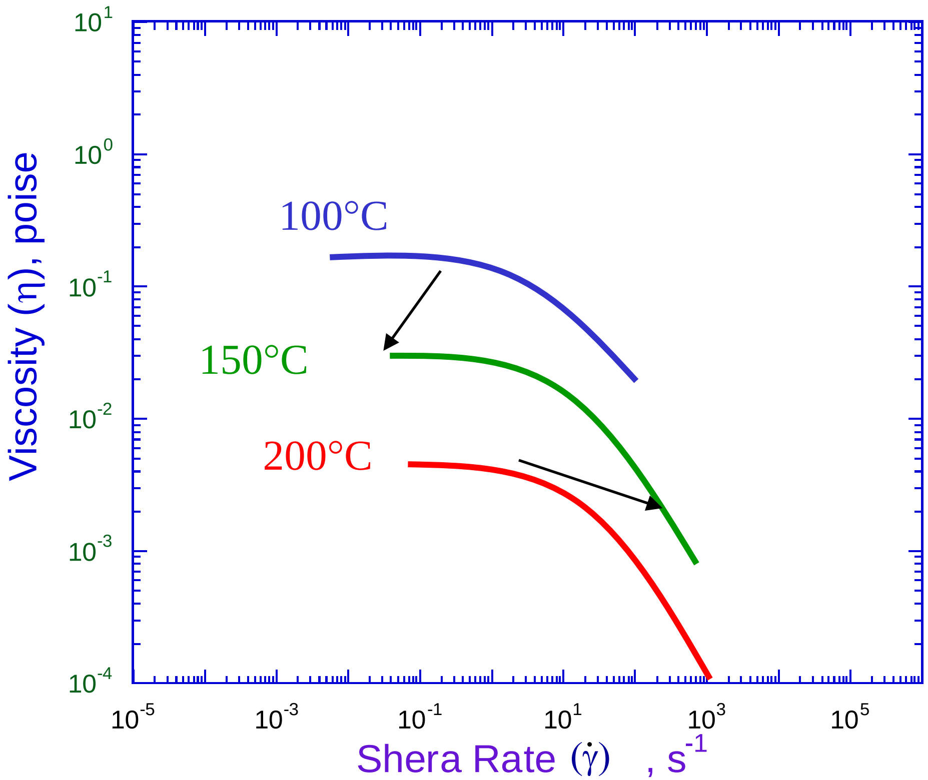
$$\eta_r \equiv reduced\ \ viscosity$$

$$= \eta(T) \quad \frac{\eta_o(T)}{\eta_o(T_{reference})}$$

TTS (Example)

Need viscosity data - polymer A @ T=150°C for shear rates 10^{-5} to 10^{6}

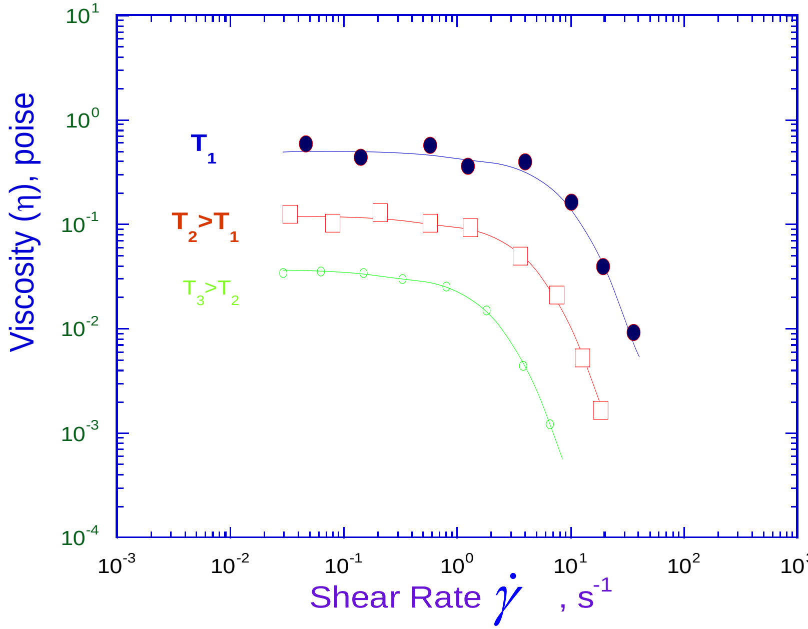
Most commercially available rheometers work in the range 10^{-3} to 10^{3} s^{-1}.

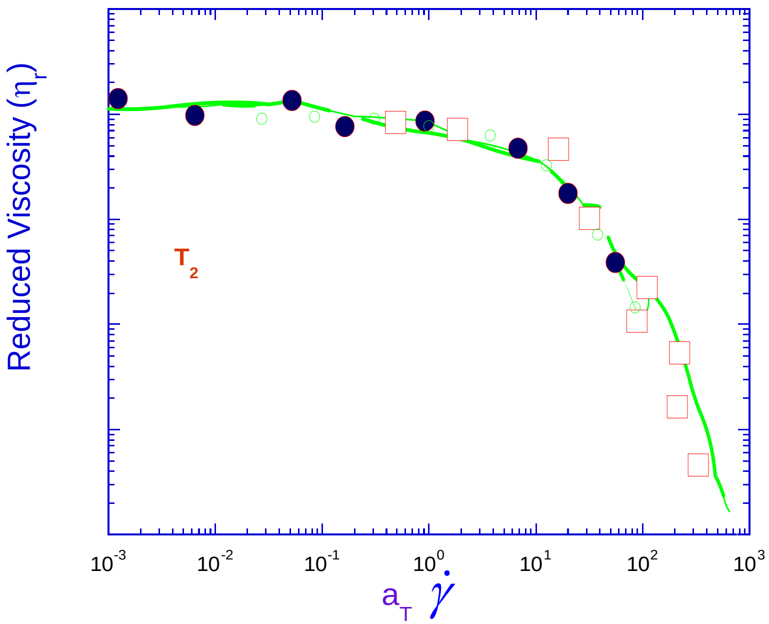


Solution:

- Measure η at different T (above and below the 150°C.
- Shift these data to T=150°C using TTS.
- Shift vertical first and then horizontally

Time-Temperature Superposition





Advantages:

Can expand shear rate range for viscosity measurements

----low shear rate data can take very long to measure

-----instrument may not have capability to take data at needed conditions

Transition!! –Dilute Solutions

So far, focus on polymer melts or concentrated solutions

Dilute Solutions:

- Solvent effect becomes more relevant
- New definitions and material function

$$\eta_{rel} = \frac{\eta}{\eta_s} = \frac{\text{Solution viscosity}}{\text{Solvent viscosity}}$$

Relative Viscosity

Dilute Solutions

$$\eta_{rel} = \frac{\eta}{\eta_s} = \frac{\text{Solution viscosity}}{\text{Solvent viscosity}}$$

$$\eta_{sp} = \frac{\eta - \eta_s}{\eta_s} = \text{specific viscosity}$$

$$\eta_{rel} = 1 + [\eta]c + K'[\eta]^2 c^2 + \ldots$$

$c \equiv$ Polymer concentration

$K' \equiv$ Huggins Coefficient

$[\eta] \equiv$ Intrinsic Viscosity

$$[\eta] = \lim_{c \to 0} \frac{\eta - \eta_s}{c\eta_s}$$

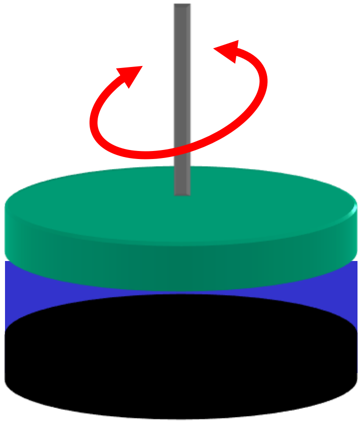
[η] related to polymer properties: coil dimension, chain stiffness

Steady Shear

- What can we get out of this measurement?
- When is this useful?
- Why do we care?

Time dependent Flow Properties

Small Amplitude Oscillatory Shear or <u>Dynamic</u> Experiment



Oscillate the top plate at a specific frequency (ω) and strain amplitude (γ_o)

Apply strain:

$$\gamma = \gamma_o \sin(\omega t)$$

$$\dot{\gamma} = \frac{d\gamma}{dt} = \gamma_o \omega \cos(\omega t)$$

$$\gamma_o \omega \equiv Shear\ Rate\ Amplitude\ \ (\dot{\gamma}_o)$$

<u>Velocity profile</u>

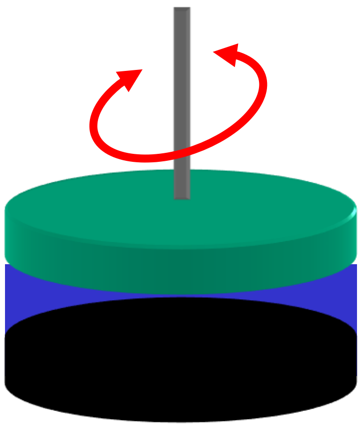
$$V_x = \dot{\gamma}_o \cos(\omega t)\, y$$

$$(V_x = \dot{\gamma}_o\, y \ in\ steady)$$

<u>Stress generated</u>

$$\tau_{yx} = A(\omega)\ \gamma_o\ \sin(\omega t + \delta)$$

Dynamic Properties



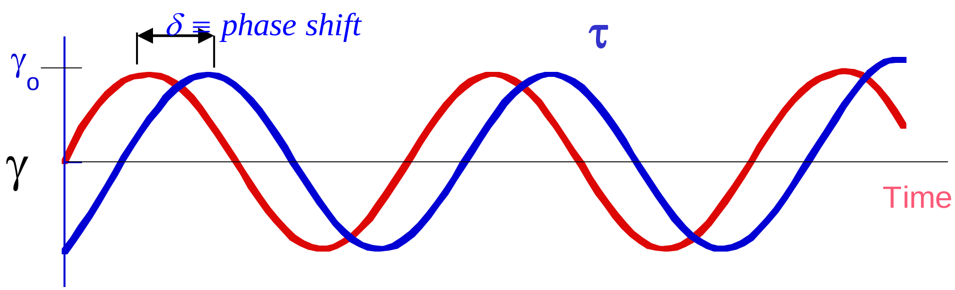
Apply strain

$\gamma = \gamma_o \sin(\omega t)$

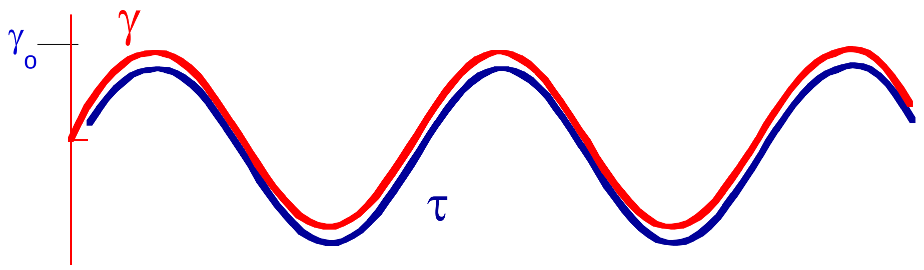
Stress generated

$\tau_{yx} = A(\omega)\, \gamma_o \, \sin(\omega t + \delta)$

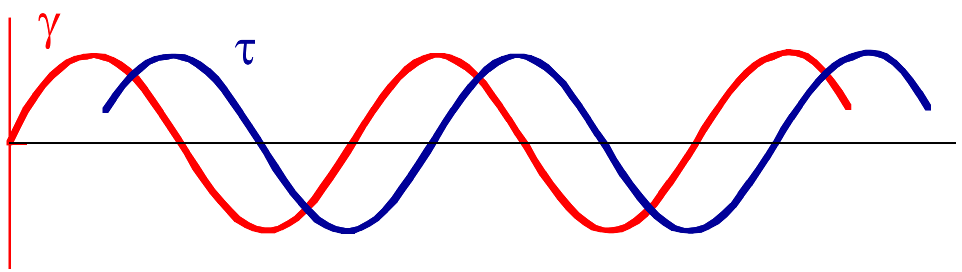
δ = *phase shift*



Dynamic Properties



Elastic solid δ=0



Viscous liquid δ=π/2

Material Functions

$$\tau_{yx} = A(\omega)\gamma_o \sin(\omega t + \delta)$$

$$\gamma = \gamma_o \sin(\omega t)$$

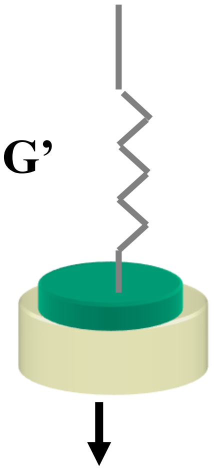
$$\tau_{yx} = G'\gamma_o \sin(\omega t) + G''\gamma_o \cos(\omega t)$$

Elastic or storage modulus

Viscous or loss modulus

Spring ~ G'

Dashpot ~ G"



For oil G'~0

For rubber G"~0

Material Functions, cont'd

G´=elastic modulus ⇒ $\eta'' \equiv \text{Dynamic Rigidity} = \dfrac{G'}{\omega}$

G´´=viscous modulus ⇒ $\eta' \equiv \text{Dynamic Viscosity} = \dfrac{G''}{\omega}$

$$\tau_{yx} = G'\gamma_o \sin(\omega t) + G''\gamma_o \cos(\omega t)$$

$$= \eta''(\omega)\dot{\gamma}_o \sin(\omega t) + \eta'(\omega)\dot{\gamma}_o \cos(\omega t)$$

$$\tan\delta \equiv \text{loss tangent} = \frac{G''}{G'}$$

$\eta^* \equiv$ Complex Viscosity

$$= \left[\eta'^2 + \eta''^2\right]^{1/2} = \left[\left(\frac{G''}{\omega}\right)^2 + \left(\frac{G'}{\omega}\right)^2\right]^{1/2}$$

Significance of Material Functions

G´=elastic modulus

Energy stored in system– shape and magnitude indicates whether sample an elastic network, gel, liquid

G´´=viscous modulus

Energy dissipated – relative magnitude w.r.t. G' indicates state of material also

$$\tan\delta \equiv \text{ loss tangent } = \frac{G''}{G'}$$

Relative importance of viscous nature over elastic nature; usually <1 gels, solids (one criteria); usually >1 viscous fluids

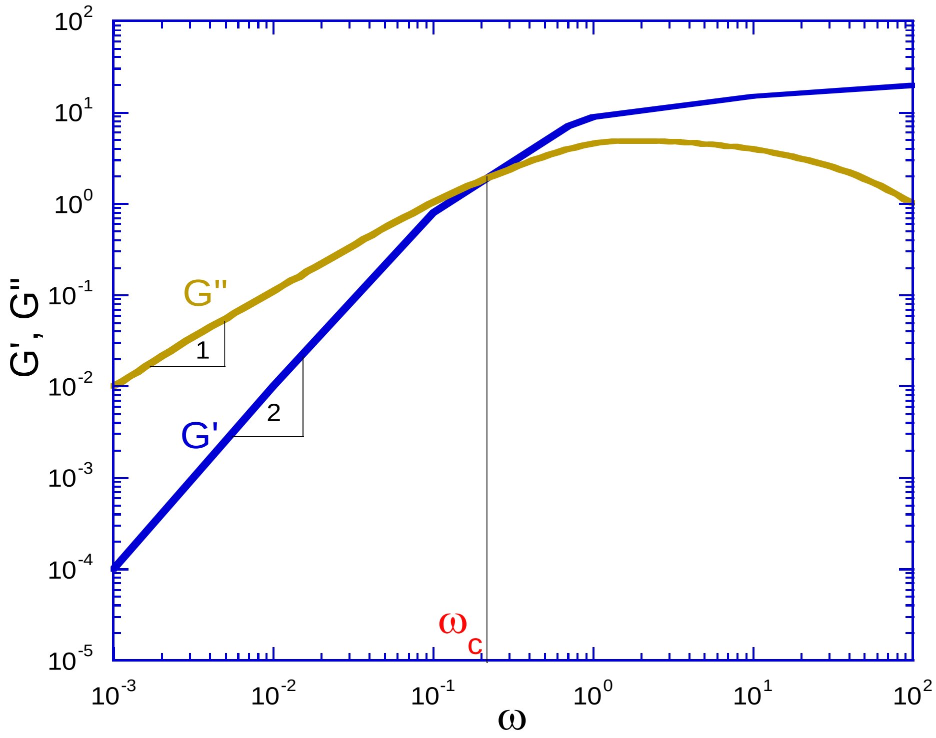
$$\eta^* \equiv \text{ Complex Viscosity}$$

$$= \left[\eta'^2 + \eta''^2\right]^{1/2} = \left[\left(\frac{G''}{\omega}\right)^2 + \left(\frac{G'}{\omega}\right)^2\right]^{1/2}$$

Analogous to steady viscosity for conventional polymeric system

Material Functions, cont'd

Typical polymer melt dynamic data



$$\omega_c = \dot{\gamma}_c$$

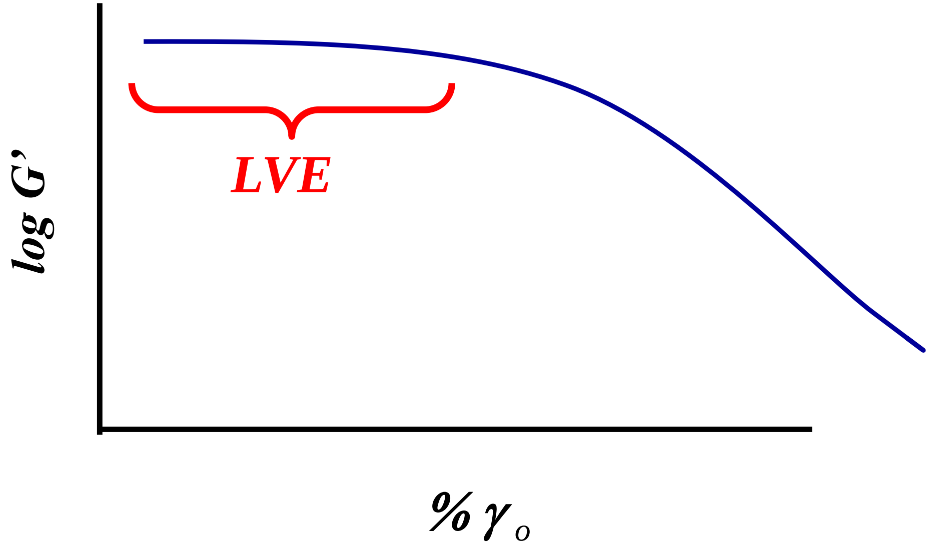
$$1/\omega_c = 1/\dot{\gamma}_c = \lambda$$

λ – longest relaxation time

$G' \sim \omega^2$; $G'' \sim \omega$ at low ω

Material Functions, cont'd

• All dynamic material functions discussed represent linear viscoelastic properties

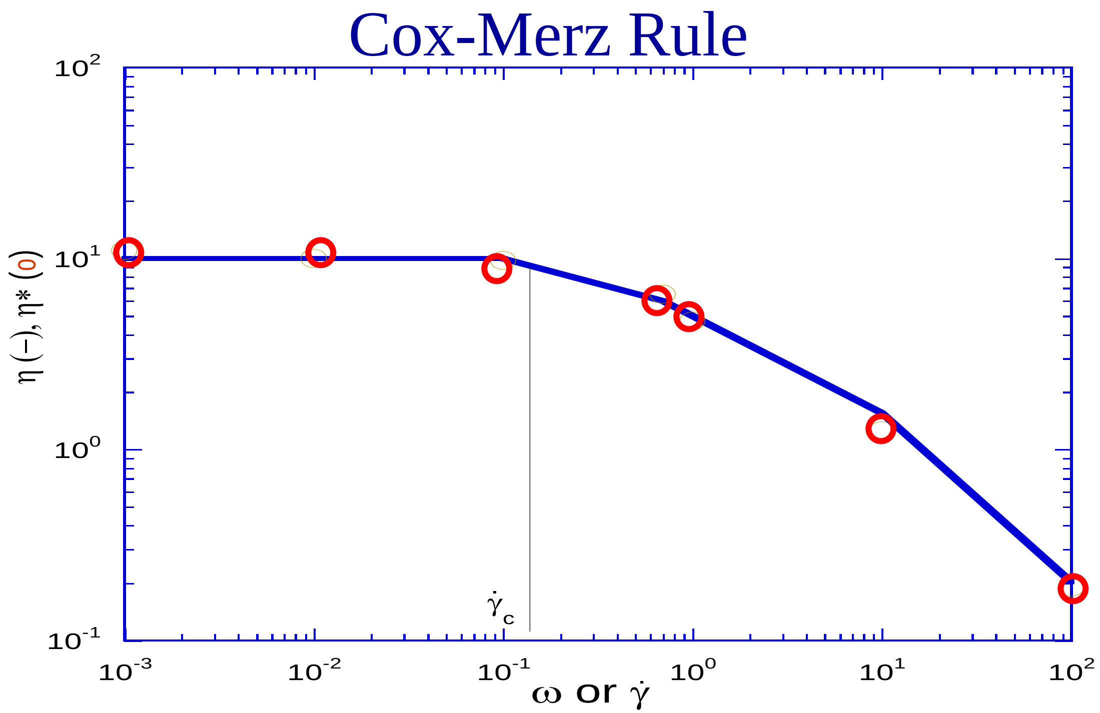


LVE **= linear viscoelastic regime**

• Experiments have to be done at v. small strain amplitude

• Do strain sweep first to get LVE; then do frequency sweep at a strain in the LVE regime

Material Functions, cont'd



$$\eta^*(\omega) = \eta(\dot{\gamma}) \qquad @ \quad \omega = \dot{\gamma}$$

$$\omega_c = \dot{\gamma}_c; \quad \frac{1}{\omega_c} \sim \frac{1}{\dot{\gamma}_c} = \lambda \qquad \lambda \equiv \textit{longest relaxation time}$$

In-class exercise

- **Plot η' and η" vs ω for a polymer**

$$\eta'' \equiv \frac{G'}{\omega}$$

$$\eta' \equiv \frac{G''}{\omega}$$

- **Are there any instances that you would prefer doing dynamic experiments over steady shear?**